INTERNSHIP REPORT

ARTIFICIAL INTELLIGENCE

Movie Recommendation System

Submitted To

School of Computer Science and Engineering
IILM University

In partial fulfilment of the requirement of degree of
B.Tech (CSE)

Submitted By

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Roll No.: 25SCS1003004111
Branch: AIML
Academic Batch: 2026–2027

Organization: Codec Technologies
Internship Domain: Artificial Intelligence
Duration: 4 Weeks

2. CANDIDATE DECLARATION

I, **Raunak Kumar**, hereby declare that this report titled “Movie Recommendation System” has been prepared for my Artificial Intelligence Internship at Codec Technologies.

The report presents the project information, methodology, technologies, learning outcomes, and conclusions based on the details supplied for the internship project.

Signature: ___________________________

Name: Raunak Kumar

Roll No.: 25SCS1003004111

3. ACKNOWLEDGEMENT

I sincerely thank Codec Technologies for providing me the opportunity to undertake an Artificial Intelligence internship and work on a practical recommendation-system project.

I am also thankful to IILM University and the School of Computer Science and Engineering for supporting practical, industry-oriented learning.

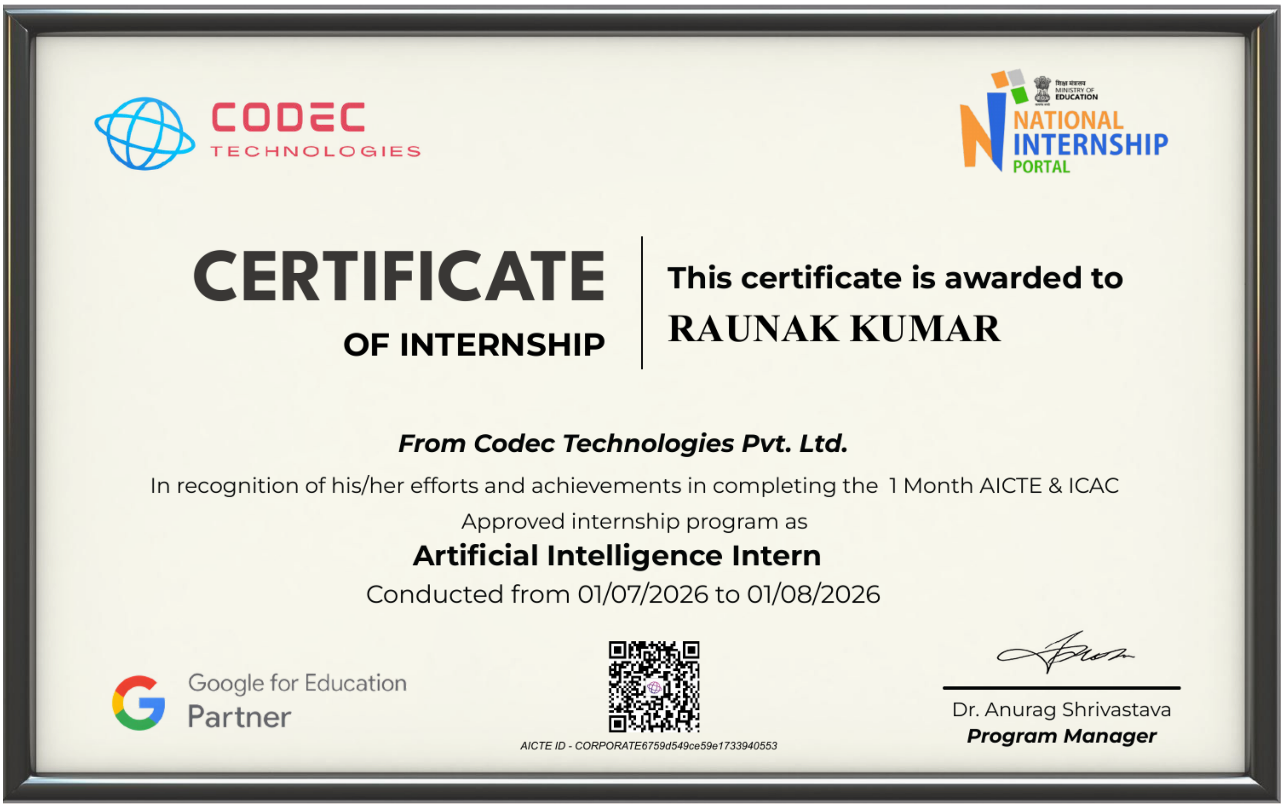
I acknowledge the use of Python, Pandas, NumPy, Scikit-Learn, NLTK, and the TMDB 5000 Movies Dataset identified for this project.

Signature: ___________________________

Raunak Kumar

4. INTERNSHIP COMPLETION CERTIFICATE

The following page reproduces the internship certificate supplied as an attachment for this report.


CODEC TECHNOLOGIES

NATIONAL INTERNSHIP PORTAL

CERTIFICATE OF INTERNSHIP

This certificate is awarded to

RAUNAK KUMAR

From Codec Technologies Pvt. Ltd.

In recognition of his/her efforts and achievements in completing the 1 Month AICTE & ICAC Approved internship program as

Artificial Intelligence Intern

Conducted from 01/07/2026 to 01/08/2026

Google for Education Partner

AICTE ID - CORPORATE6759d549ce59e1733940553

Dr. Anurag Shrivastava

Program Manager

5. TABLE OF CONTENTS

1. INTRODUCTION

The Artificial Intelligence Internship at Codec Technologies provided practical exposure to data processing and recommendation-system concepts. The project selected for the internship is a **Movie Recommendation System**.

The workflow starts with the TMDB 5000 Movies Dataset and proceeds through dataset understanding, preprocessing, feature engineering, text cleaning, vectorization, similarity calculation, and a recommendation function.

The system is designed to help users discover movies that match their interests by using movie attributes and, where available, ratings, reviews, watch history, feedback, and collaborative patterns.

The project demonstrates how Python and machine-learning libraries can be applied to a practical Artificial Intelligence problem.

2. ORGANIZATION PROFILE

Codec Technologies is the organization identified for the Artificial Intelligence Internship.

Internship Details

- Organization: Codec Technologies
- Domain: Artificial Intelligence Internship
- Duration: 4 Weeks
- AICTE Student ID: STU6a906a79314141787849337
- APAAR ID: 988844650968
- Certificate / Corporate AICTE ID: CORPORATE6759d549ce59e1733940553
- Offer Letter Reference: E19E86-0116588288923

The internship project focused on applying data-processing, natural-language processing, and recommendation techniques to a movie dataset.

3. INTERNSHIP OVERVIEW AND PROBLEM STATEMENT

Project: Movie Recommendation System

Problem Statement

Movie catalogues can contain a large number of titles, making it difficult for users to quickly find suitable movies. A recommendation system can reduce this search effort by analysing movie characteristics and user-related signals.

The project aims to build a simple recommendation system that suggests movies based on user ratings or genres and demonstrates content-based or collaborative filtering techniques.

Main Project Features

- Genre and tags
- Cast and crew
- Movie descriptions
- Metadata
- Watch history and habits
- Ratings and reviews
- User feedback
- Collaborative patterns

4. PROJECT OBJECTIVES AND SCOPE

Objectives

1. Build a simple movie recommendation system.
2. Use the TMDB 5000 Movies Dataset (Kaggle).
3. Perform preprocessing and feature engineering.
4. Clean and prepare text information.
5. Apply TF-IDF or Count Vectorizer for text representation.
6. Use Cosine Similarity for movie comparison.
7. Understand content-based and collaborative filtering.
8. Understand popularity-based and hybrid approaches.
9. Generate movie suggestions through a recommendation function.

Scope

The scope covers movie genres, tags, cast, crew, descriptions, metadata, ratings, reviews, feedback, watch history, and collaborative patterns. The supplied information does not specify production deployment or measured accuracy, so these are not claimed as completed deliverables.

5. TECHNOLOGIES AND TOOLS

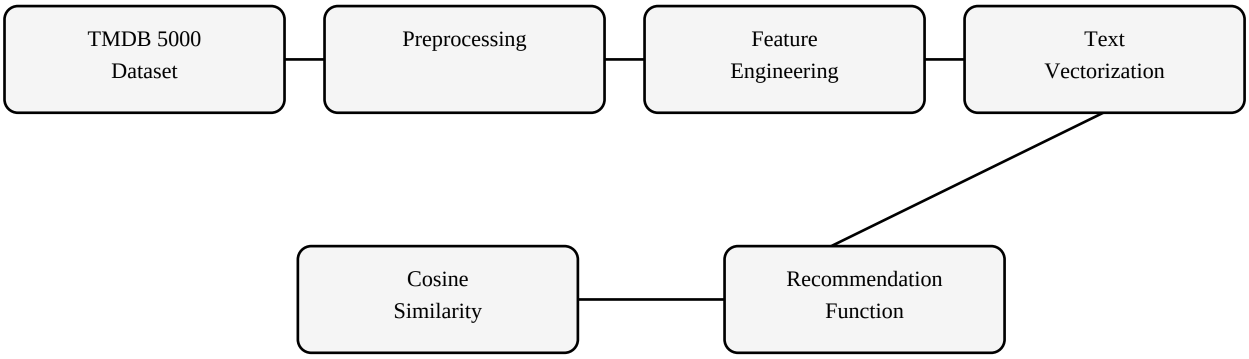
The project uses Python and the following libraries and technologies:

Technology	Application
Python	Primary programming language
Pandas	Data loading, analysis and preprocessing
NumPy	Numerical operations
Scikit-Learn	Vectorization and similarity utilities
NLTK	Natural-language text cleaning
TMDB 5000 Movies Dataset	Movie dataset sourced through Kaggle
TF-IDF / Count Vectorizer	Text vectorization
Cosine Similarity	Similarity measurement

Technical sequence: Dataset overview → Data columns → Preprocessing → Text cleaning → Feature engineering → Vectorization → Recommendation.

6. SYSTEM ARCHITECTURE AND METHODOLOGY

The recommendation pipeline converts raw movie information into useful features and then compares movie representations to produce suggestions.



Methodology

Content-Based Filtering: Recommends movies similar to titles a user likes by analysing genre, cast, director, plot tags, and related features. TF-IDF/Count Vectorizer and Cosine Similarity can be used for feature comparison.

Collaborative Filtering: Analyses ratings, reviews, watch history, and similar-user behaviour to identify recommendation patterns.

Popularity Filtering: Suggests globally popular or highly rated movies as a baseline, especially for users with limited history.

Hybrid Systems: Combine content-based and collaborative signals to reduce individual limitations such as cold-start situations.

7. IMPLEMENTATION AND RESULTS

1. Dataset Overview: The TMDB 5000 Movies Dataset (Kaggle) is used as the project data source.

2. Data Columns: Relevant movie information is identified for use in preprocessing and feature engineering.

3. Preprocessing: Data is prepared and cleaned before feature extraction.

4. Text Cleaning: Movie-related textual information is cleaned for subsequent vectorization; NLTK is included among the project libraries.

5. Feature Engineering: Genre, tags, cast, crew, descriptions, and metadata are organised into useful recommendation features.

6. Vectorization: TF-IDF or Count Vectorizer represents text information numerically.

7. Similarity: Cosine Similarity measures similarity between movie vectors.

8. Recommendation: The recommendation function outputs movie suggestions.

Result: The workflow provides a structured method for generating relevant movie suggestions. No numerical accuracy or performance metric was supplied, so no metric is claimed.

8. LEARNING OUTCOMES, CHALLENGES AND CONCLUSION

Learning Outcomes

- Understanding of the complete movie-recommendation workflow.
- Practical experience with dataset inspection and preprocessing.
- Understanding of feature engineering and text cleaning.
- Understanding of TF-IDF/Count Vectorizer and Cosine Similarity.
- Understanding of content-based, collaborative, popularity-based, and hybrid recommendation approaches.
- Experience using Python with Pandas, NumPy, Scikit-Learn, and NLTK.

Challenges and Solutions

- Large catalogue → use feature-based similarity.
- Textual information → clean and vectorize text.
- Different preferences → use movie attributes and behavioural signals when available.
- Cold-start limitation → use popularity-based baselines and hybrid approaches.

Conclusion

The Movie Recommendation System demonstrates an end-to-end Artificial Intelligence workflow from dataset preparation to recommendation output. It shows how movie metadata and user-related signals can be used with Python-based tools to support movie discovery.

9. FUTURE SCOPE, REFERENCES AND ANNEXURE

Future Scope

- Develop a stronger hybrid recommendation engine.
- Improve user-profile and feedback handling.
- Explore additional machine-learning or deep-learning recommendation models.
- Improve cold-start handling for new users and movies.
- Evaluate recommendation quality with appropriate metrics.
- Develop an interactive application for movie discovery.

References

1. TMDB 5000 Movies Dataset, Kaggle.
2. Scikit-Learn documentation.
3. NLTK documentation.
4. Pandas documentation.
5. NumPy documentation.

Annexure — Student & Project Information

Raunak Kumar | 25SCS1003004111 | B.Tech (CSE) | AIML | 2026–2027 | IILM University
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